

Multiple Choice (10 marks)

1. A proxy rewall lters at the
 - (a) Physical layer
 - (b) Application layer
 - (c) Data link layer
 - (d) Network layer
2. Three of the following are classic security properties; which one is not?
 - (a) Confidentiality
 - (b) Availability
 - (c) Correctness
 - (d) Integrity
3. Which of the following is not a security exploit?
 - (a) Eavesdropping
 - (b) Cross-site scripting
 - (c) Authentication
 - (d) SQL Injection
4. Of the following, which best approximates the ratio of the number of nonterminal nodes to the total number of nodes in a complete K-ary tree of depth N?
 - (a) $1/K$
 - (b) $K-1/K$
 - (c) $\log_{10} (1/N)$
 - (d) $N-1/m$
5. _____ is a widespread app's coding mistake made by developers which could be exploited by an attacker for gaining access or malfunctioning your system.
 - (a) Memory leakage
 - (b) Buffer-overflow
 - (c) Less processing power
 - (d) Inefficient programming

True / False (10 marks)

Answer True or False

6. True or False: Wireless security involves measures taken to protect computers and data from unauthorized access via wireless networks.
7. True or False: Church's thesis was first proven by Alan Turing and is widely accepted as a formal theorem in computability theory.
8. True or False: WPS is a protocol designed to simplify the process of connecting new devices to wireless access points.
9. True or False: The set of Boolean operators {AND, OR} is not complete for representing all Boolean expressions.
10. True or False: A backdoor is a method that allows unauthorized access to a computer system while bypassing security mechanisms.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to create a list taking alternate elements from another given list.
12. Write a function to sort a given mixed list of integers and strings.

End of Paper